

4 - BIT CARRY LOOK AHEAD ADDER

Abstract—This project presents a high-speed, low-power 4-bit carry-lookahead adder (CLA) using optimized static CMOS logic. The design reduces critical path delay by 15-20

■ **THE AIM OF THIS PROJECT** is to design a 4-bit carry-lookahead adder (CLA) that computes the sum and output carry for two 4-bit inputs and an input carry. The circuit operates on the rising clock edge and drives inverters sized at $W_p/W_n = 20\lambda/10\lambda$, with $\lambda = 0.09 \mu\text{m}$. The design minimizes the power-delay product (PDP) using optimized topologies.

Introduction

Adders are fundamental in processors and arithmetic operations, making their speed critical for complex logic implementations. While ripple carry adders are compact, their feedback-based carry propagation causes delays. The carry-lookahead adder addresses this by calculating carries simultaneously, significantly reducing delay and PDP. This project enhances the conventional CLA design, achieving lower delay with a minor increase in area and power, making it suitable for modern VLSI systems.

Proposed CLA Circuit

The circuit shown below illustrates a conventional CLA design, highlighting its longest delay path. The fundamental equations governing the CLA are as follows:

$$P_i = A_i \oplus B_i \quad (1)$$

$$G_i = A_i \cdot B_i \quad (2)$$

$$C_i = G_{i-1} + P_{i-1}C_{i-1} \quad (3)$$

$$S_i = P_i \oplus C_i \quad i = 1, 2, 3, 4 \quad (4)$$

Using the above equations, we precompute the expressions for the carry at each stage. The derived equations are as follows:

$$\begin{aligned} C_1 &= G_0 + P_0C_0 \\ C_2 &= G_1 + P_1C_1 = G_1 + P_1G_0 + P_1P_0C_0 \\ C_3 &= G_2 + P_2C_2 = G_2 + P_2G_1 + P_2P_1G_0 + P_2P_1P_0C_0 \\ C_4 &= G_3 + P_3C_3 = G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0 + P_3P_2P_1P_0C_0 \end{aligned}$$

The total critical path delay of the CLA can be expressed as:

$$\text{Delay} : D_{OR-5} + D_{AND-5} + D_{XOR-2} \quad (5)$$

The 5-input AND and OR gates are the primary contributors to the larger delay. To address this, we propose the following modified topology.

The proposed topology breaks the carry at every stage into two parts, i.e.,

$$C_i = X_i + Y_i \cdot C_0 \quad (6)$$

The equation for C_1 is already in the required form since

$$C_1 = G_0 + P_0 \cdot C_0 \quad (7)$$

The equations for C_2 , C_3 , and C_4 can be broken down using this logic as follows:

For C_2 :

$$X_2 = G_1 + P_1 G_0$$

$$Y_2 = P_1 P_0 \cdot C_0$$

For C_3 :

$$X_3 = G_2 + P_2 G_1 + P_2 P_1 G_0$$

$$Y_3 = P_2 P_1 P_0 \cdot C_0$$

For C_4 :

$$X_4 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0$$

$$Y_4 = P_3 P_2 P_1 P_0 \cdot C_0$$

The function $AB + C$ is implemented using gates called AO21. These gates require 8 transistors when implemented using static CMOS logic. When combined in the circuit, they offer better delay performance compared to the five-input gates used in a conventional CLA module.

As shown in the circuit diagram, the total critical path delay of the CLA can be expressed as:

$$Delay : D_{AO21} + D_{OR-4} + D_{AND-4} + D_{XOR-2} \quad (8)$$

This hypothesis has been verified by constructing a conventional CLA model and comparing the delays of the two circuits using NGSPICE. The details of the results are discussed in subsequent sections of the report.

Subcircuit Block Specifications

The modified clocked CLA adder incorporates various circuit blocks to achieve full functionality. It was observed that the minimum delay is achieved when sizing an inverter with W_p/W_n as $20\lambda/10\lambda$.

When the size was reduced to $20\lambda/10\lambda$, there was a $30ps$ increase in the critical path delay. However, this reduction in size resulted in a significant decrease in area. Consequently, all gates were sized accordingly to minimize delay, while comparing their performance to an inverter with W_p/W_n as $20\lambda/10\lambda$.

To achieve a good power-delay product (PDP), static CMOS logic was exclusively used to construct the logic gates. While implementing the gates with dynamic logic could reduce delay, it would significantly increase power dissipation. A comparison of the

trade-offs indicated that static CMOS logic provided a lower PDP compared to dynamic logic, making it the preferred choice for this design.

Flip Flops

During the implementation of flip flops, two novel architectures were considered: True Single Phase Clocked Logic (TSPC) and Current Mode Logic (CML). Upon analysis, it was observed that CML provides a lower delay compared to TSPC.

However, this advantage comes at the cost of a significant increase in power dissipation. Additionally, TSPC is easier to implement and requires less area due to the reduced number of transistors compared to CML. For these reasons, TSPC was ultimately chosen for the design.

Subcircuits

$W = 10\lambda$

• 2-Input AND Gate:

- **Sizing:** PMOS: $2W$, NMOS: W
- **Topology:** Static CMOS Logic

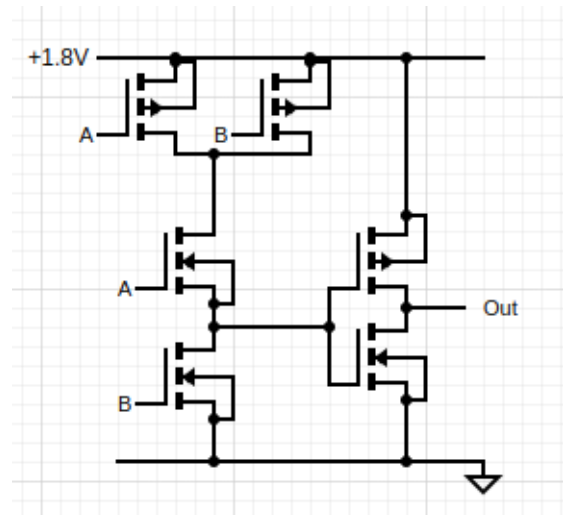


FIGURE 1. 2-Input AND Gate

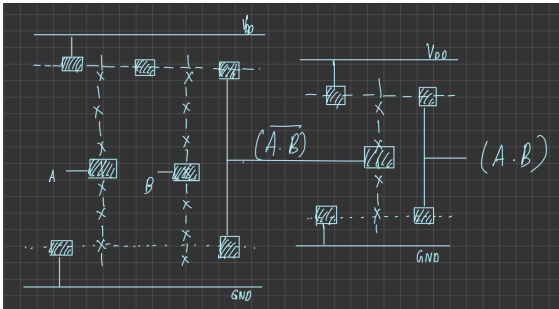


FIGURE 2. 2 INPUT AND STICK DIAGRAM

• 3-Input AND Gate:

- Sizing: PMOS: $2W$, NMOS: W
- Topology: Static CMOS Logic

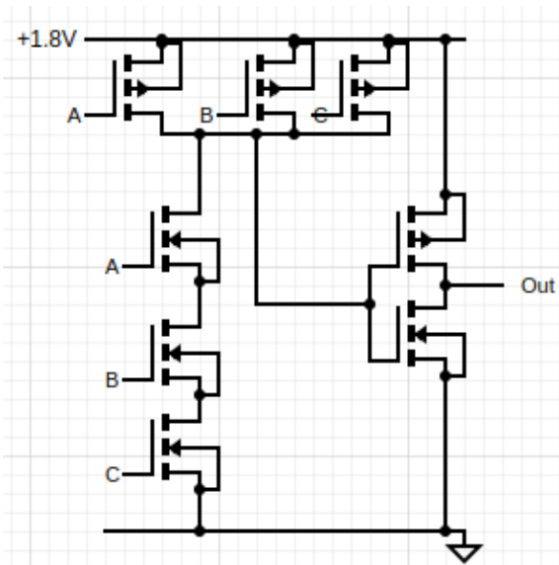


FIGURE 3. 3 INPUT AND GATE

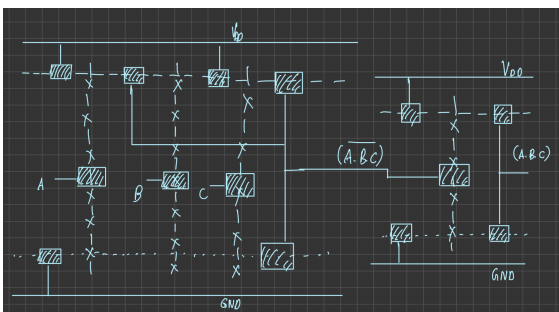


FIGURE 4. 3 INPUT AND STICK DIAGRAM

• 4-Input AND Gate:

- Sizing: PMOS: $2W$, NMOS: W

- Topology: Static CMOS Logic

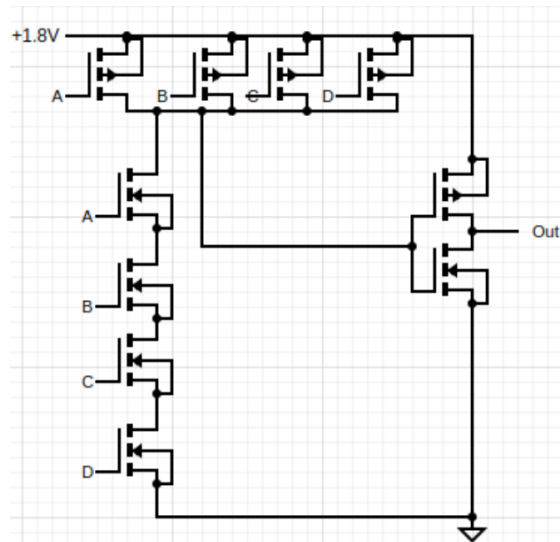


FIGURE 5. 4 INPUT AND GATE

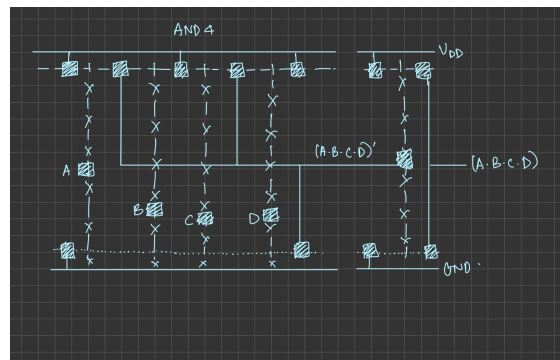


FIGURE 6. 4 INPUT STICK DIAGRAM

• 5-Input AND Gate:

- Sizing: PMOS: $2W$, NMOS: W
- Topology: Static CMOS Logic

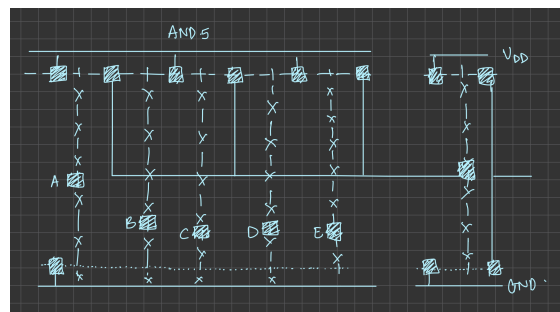


FIGURE 7. 5 INPUT AND STICK DIAGRAM

- **2-Input OR Gate:**

- **Sizing:** PMOS: $2W$, NMOS: W
- **Topology:** Static CMOS Logic

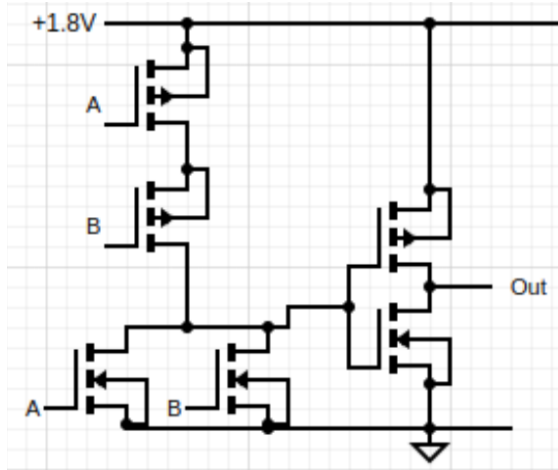


FIGURE 8. 2 INPUT OR GATE

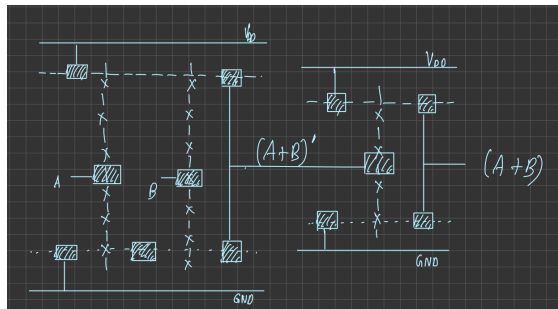


FIGURE 9. 2 INPUT OR GATE STICK DIAGRAM

- **3-Input OR Gate:**

- **Sizing:** PMOS: $2W$, NMOS: W
- **Topology:** Static CMOS Logic

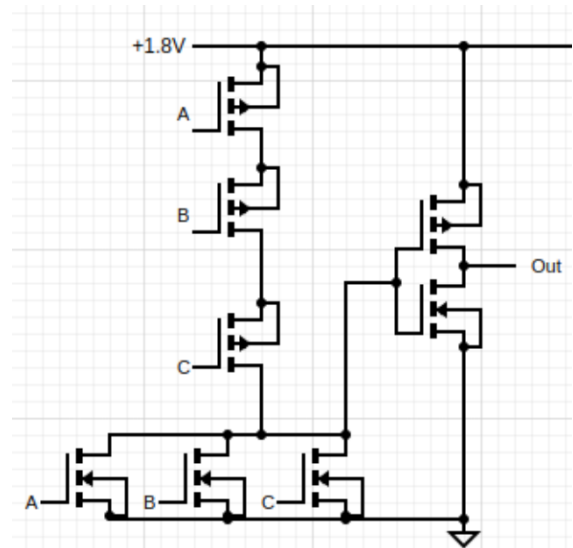


FIGURE 10. 3 INPUT OR GATE

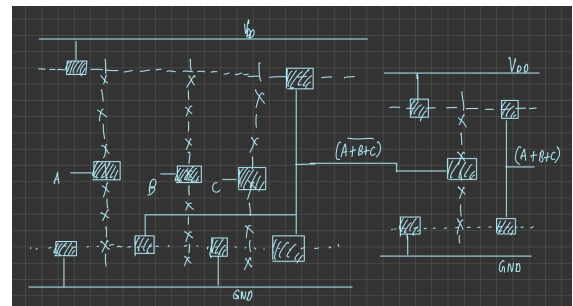


FIGURE 11. 3 INPUT OR GATE STICK DIAGRAM

- **4-Input OR Gate:**

- **Sizing:** PMOS: $2W$, NMOS: W
- **Topology:** Static CMOS Logic

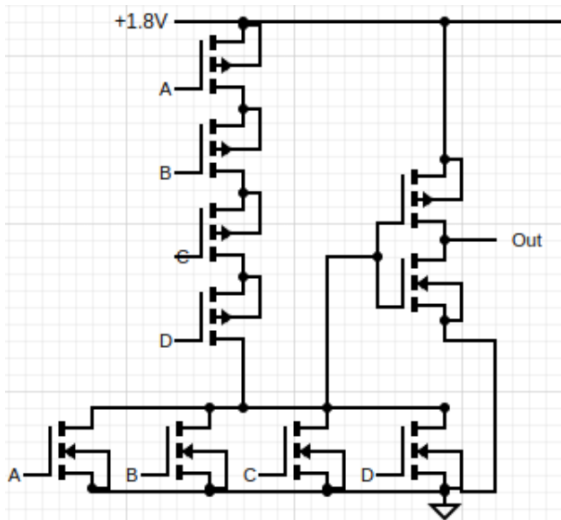


FIGURE 12. 4 INPUT OR GATE

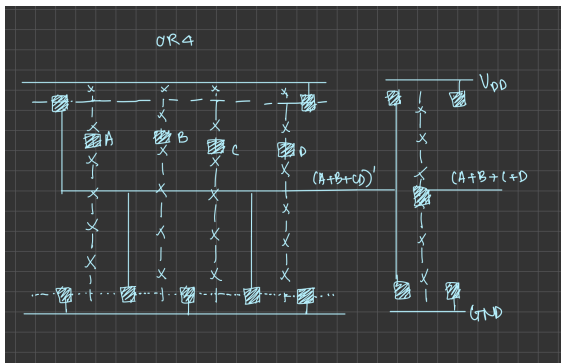


FIGURE 13. 4 INPUT OR GATE STICK DIAGRAM

• 5-Input OR Gate:

- Sizing: PMOS: $2W$, NMOS: W
- Topology: Static CMOS Logic

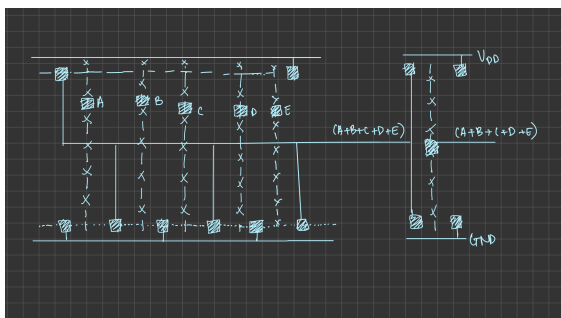


FIGURE 14. 5 INPUT AND GATE STICK DIAGRAM

• 2-Input XOR Gate:

- Sizing: PMOS: $2W$, NMOS: W

- Topology: Static CMOS Logic

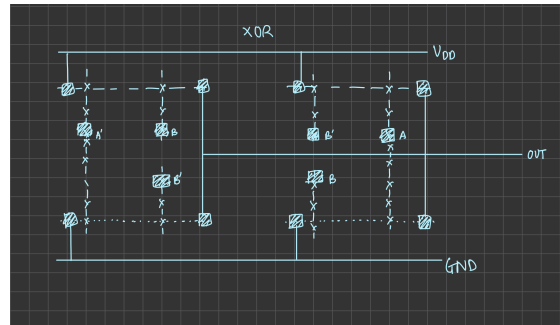


FIGURE 15. 2 INPUT XOR GATE STICK DIAGRAM

• TSPC FLIP FLOP:

- Sizing: PMOS: W , NMOS: W
- Topology: Static CMOS Logic

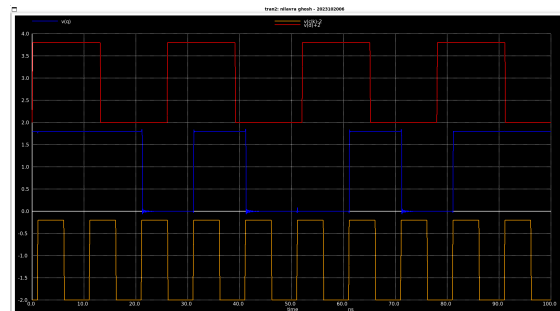


FIGURE 16. OUTPUT INPUT TSPC FLIP FLOPS

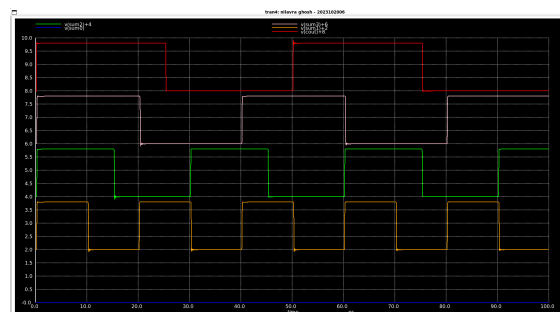


FIGURE 17. NGSPICE CLA PRE-LAYOUT

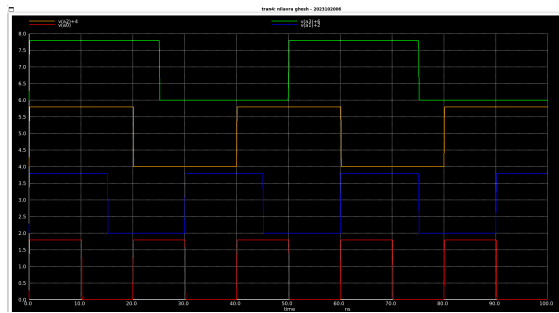


FIGURE 18. NGSPICE CLA PRE-LAYOUT INPUT i.e BOTH ARE SAME

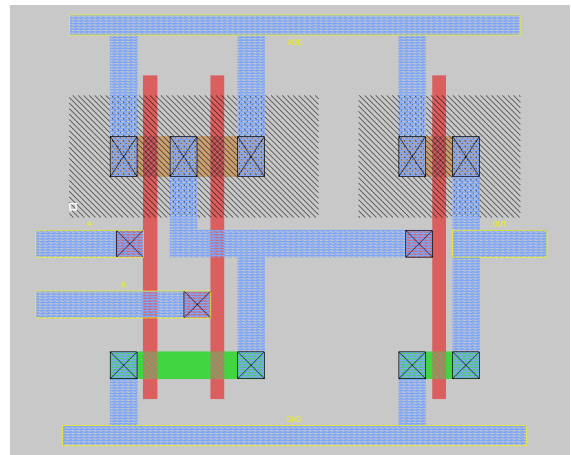


FIGURE 21. 2 INPUT AND GATE MAGIC LAYOUT

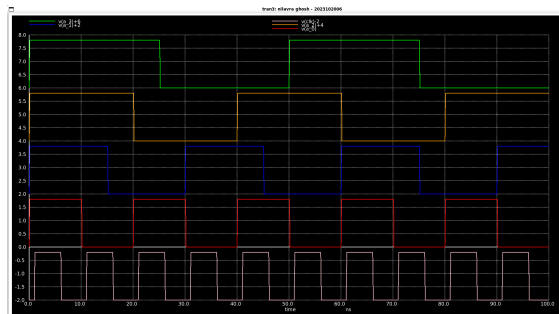


FIGURE 19. NGSPICE INPUT i.e BOTH INPUT ARE SAME

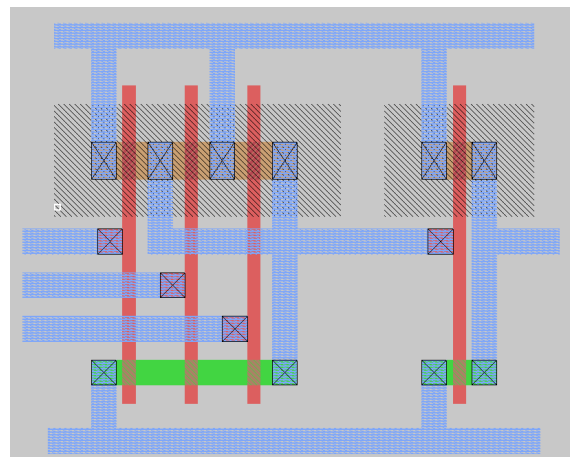


FIGURE 22. 3 INPUT AND GATE MAGIC LAYOUT

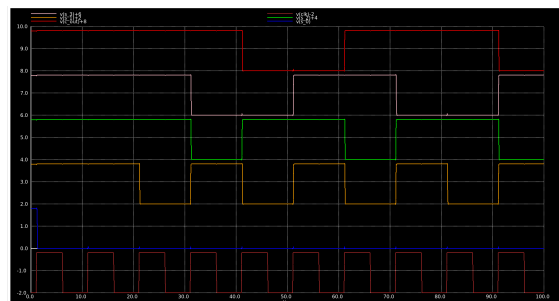


FIGURE 20. NGSPICE OUTPUT

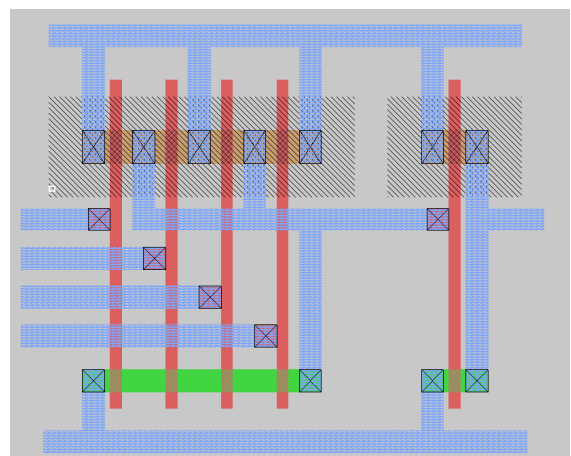


FIGURE 23. 4 INPUT AND GATE MAGIC LAYOUT

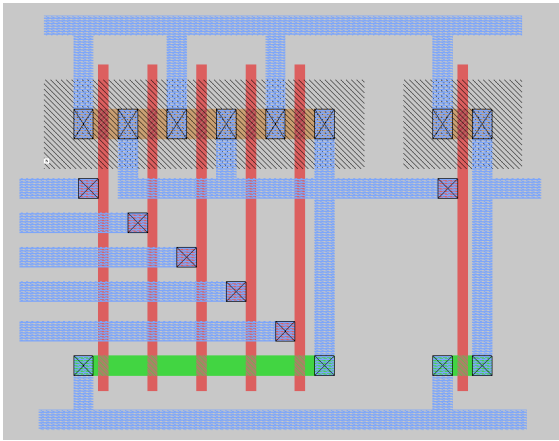


FIGURE 24. 5 INPUT AND GATE MAGIC LAYOUT

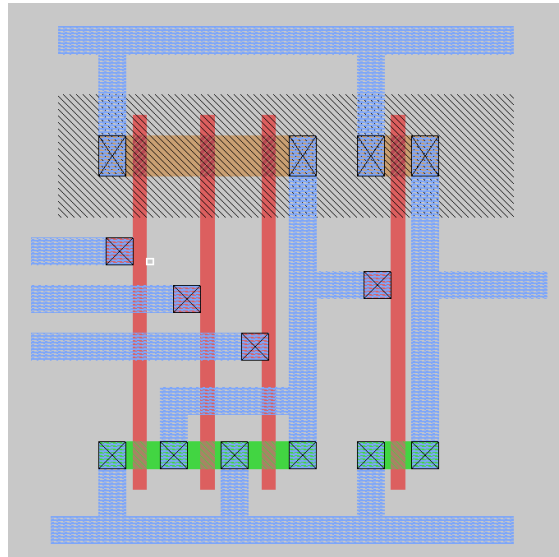


FIGURE 26. 3 INPUT OR GATE MAGIC LAYOUT

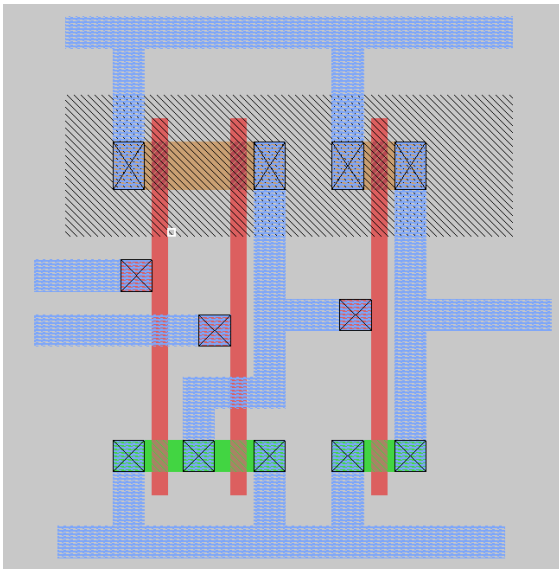


FIGURE 25. 2 INPUT OR GATE MAGIC LAYOUT

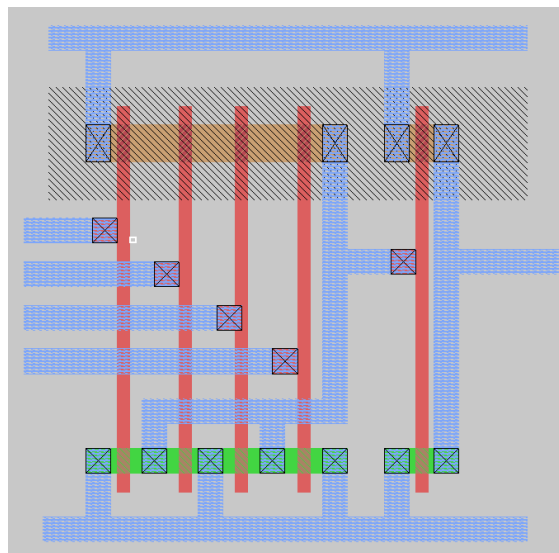


FIGURE 27. 4 INPUT OR GATE MAGIC LAYOUT

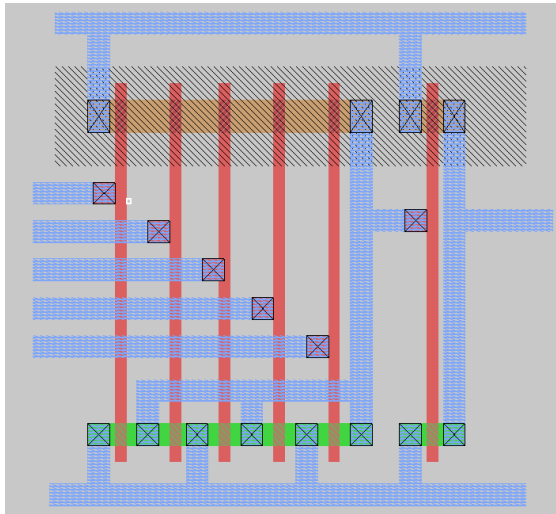


FIGURE 28. 5 INPUT OR GATE MAGIC LAYOUT

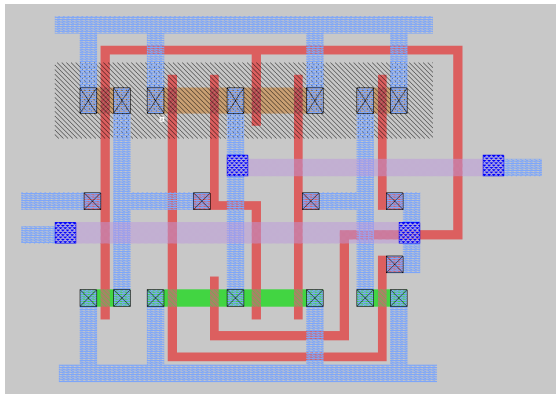


FIGURE 29. XOR GATE MAGIC LAYOUT

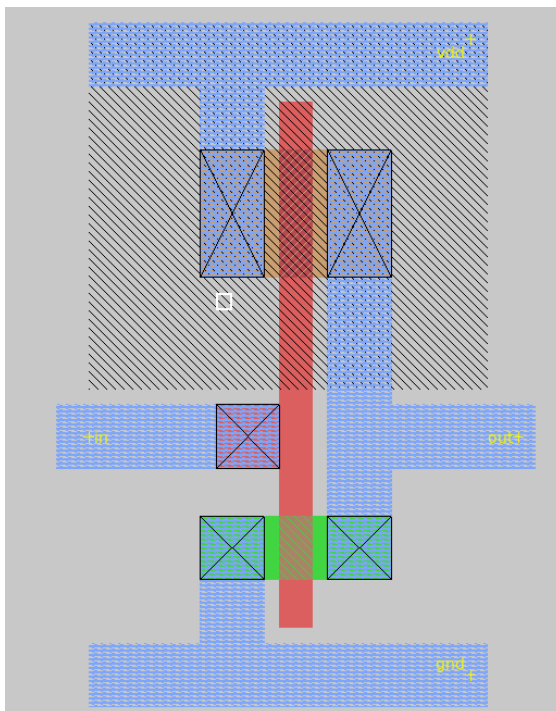


FIGURE 30. INVERTER MAGIC LAYOUT

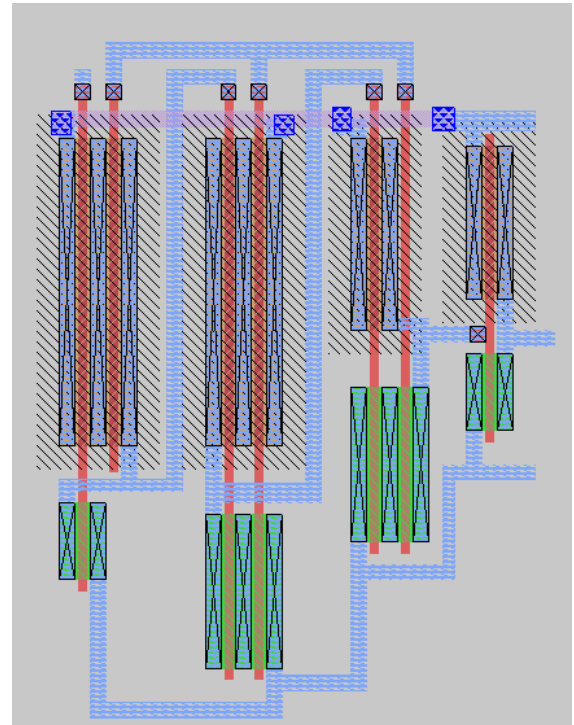


FIGURE 31. TSPC FLIP FLOP MAGIC LAYOUT

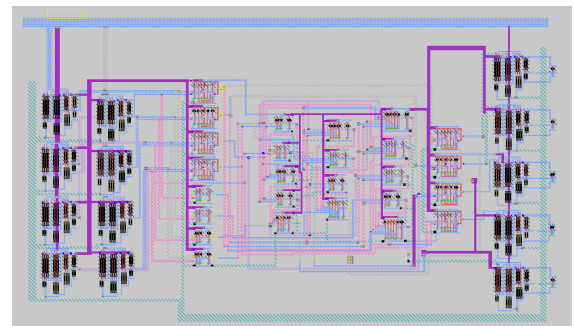


FIGURE 32. 4-BIT CLA ADDER WITH TSPC FLIP FLOPS AND INVERTERS ADDED AT LAST



FIGURE 33. VERILOG GTK WAVE PLOT

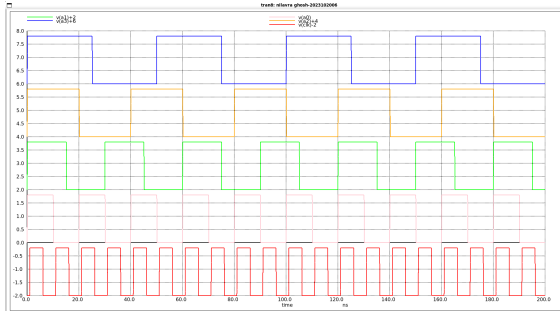


FIGURE 34. POST LAYOUT INPUTS WHERE BOTH INPUTS ARE THE SAME

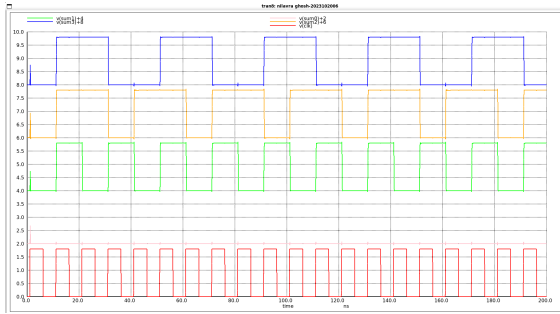


FIGURE 35. POST LAYOUT FINAL CIRCUIT OUTPUT SUM

```
tpcq_in      = 5.900000e-08 targ= 6.005000e-08 trig= 1.050000e-09
tpd          = 3.116411e-08 targ= 3.121411e-08 trig= 5.000000e-11
```

FIGURE 36. PRE LAYOUT PROPAGATION DELAY AND TPCQ

```
tpcq_in      = 5.900000e-08 targ= 6.005000e-08 trig= 1.050000e-09
tpd          = 2.146308e-08 targ= 2.151308e-08 trig= 5.000000e-11
```

FIGURE 37. POST LAYOUT PROPAGATION DELAY AND TPCQ

Given Parameters:

- $t_{pd} = 214 \text{ ps}$
- $t_{pcq} = 590 \text{ ps}$
- $t_{su} = 60 \text{ ps}$

A stable output is obtained only if the following expression holds:

$$T_{Clk} \geq t_{pd} + t_{pcq} + t_{su}$$

$$\Rightarrow T_{Clk} \geq 0.864 \text{ ns}$$

Therefore, the maximum operable clock frequency is:

$$f_{max} = \frac{1}{T_{Clk,min}} = 1.15 \text{ GHz}.$$

However, the provided netlist includes a certain error margin, with the clock period set to 2 ns, implying a clock frequency of:

$$f_{netlist} = \frac{1}{2 \text{ ns}} = 0.5 \text{ GHz}.$$